

Gamifying the break from routine

MiniQuest is a native Android RPG that turns response interruption and environmental enrichment into a playable system, using phone sensors to reward movement beyond habitual routines.

Designer & developer

Shuhan Zhang

Module

Mobile Interaction

Platform

Android · Kotlin

Role

Individual project



WATCH

Test video

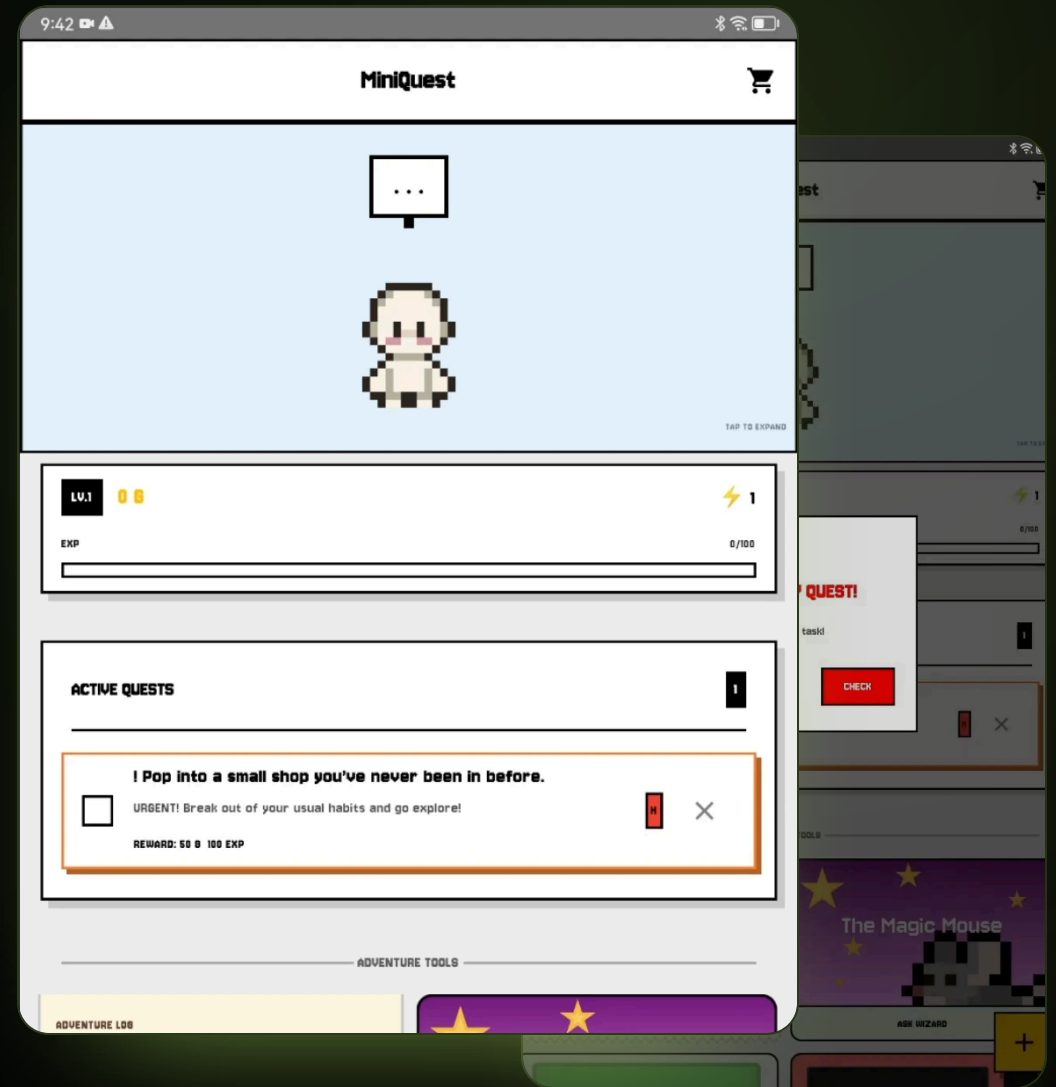
youtube.com/shorts/CBCdqMNU3ZE



EXPLORE

Project GitHub

github.com/Shuhan0001/MiniQuest



Breaking the rigidity loop

Behavioral research reframed repetitive routine as an interaction problem: the app must interrupt the loop, then offer enough novelty to prevent an immediate return.

01

Monotony reinforces repetition

Under-stimulating environments are associated with more stereotyped behavior; routine can become its own closed environment.

Design implication: reward variation, not only completion.

02

Response interruption

A well-timed alternative action can redirect attention before a repetitive pattern continues unchecked.

Design response: emergency quests surface without being pre-planned.

03

Environmental enrichment

Greater sensory and activity variety gives users more routes away from a narrow behavioral loop.

Design response: multiple adventure and regulation tools.

A conventional habit tracker rewards staying inside the loop. MiniQuest rewards stepping outside it.

Research context

Stereotyped behaviour is associated with ASD but also appears, at lower frequency and intensity, in adults without ASD. The design translates response interruption and environmental enrichment into distinct interaction mechanics.

Rapp & Vollmer, 2005; Nwaordu & Charlton, 2024; Gross et al., 2012; Turner, Yang & Lewis, 2002; Boyd, McDonough & Bodfish, 2012.

Two routes beyond routine

The RPG reward loop is retained, but progress no longer depends entirely on tasks the user already planned.

01 / INTERRUPT

Urgent quests

The system introduces an unexpected action at the moment routine needs to be broken, turning redirection into a short playable challenge.

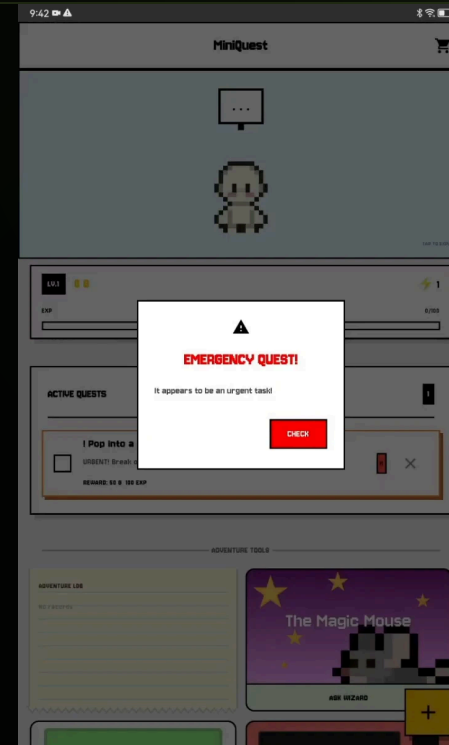
02 / ENRICH

Adventure tools

Stress relief, exploration and passive movement rewards create several low-friction ways to re-enter the world beyond the to-do list.

BEYOND HABITICA

Passive sensing rewards walking in the background. **Induced novelty** lets the system, not the user, decide when a new quest appears.



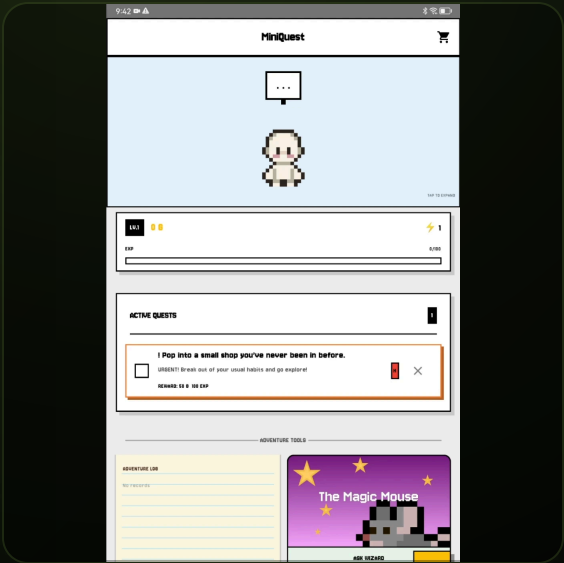
Induced novelty



Sensory regulation

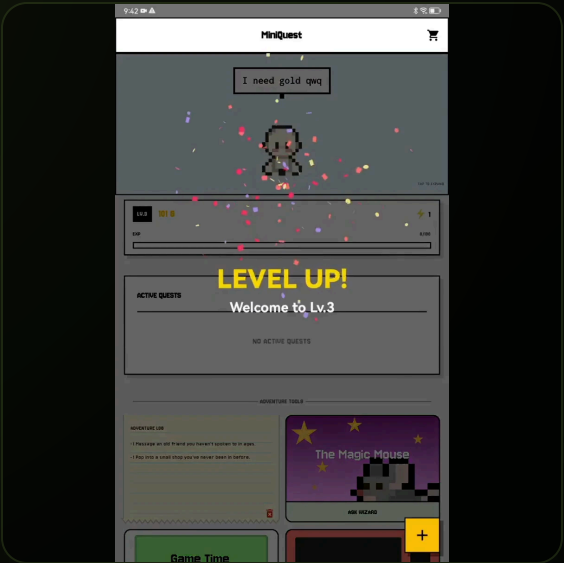
A pocket-sized adventure loop

One visual system carries the user from everyday quests to celebration, guidance and reward redemption.



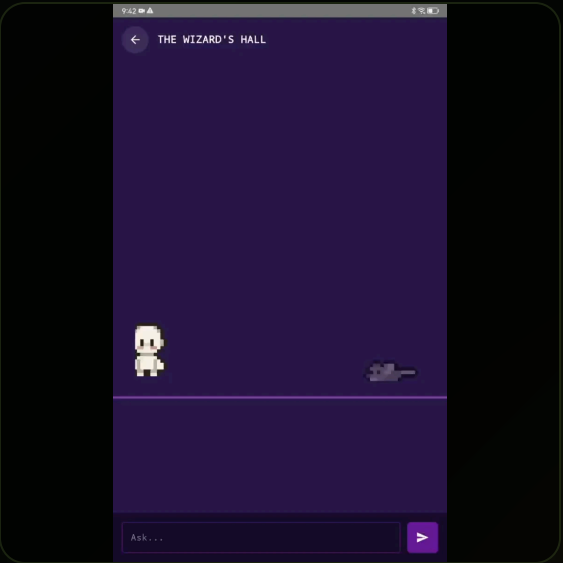
01 Quest log

Daily progress, stats and passive rewards.



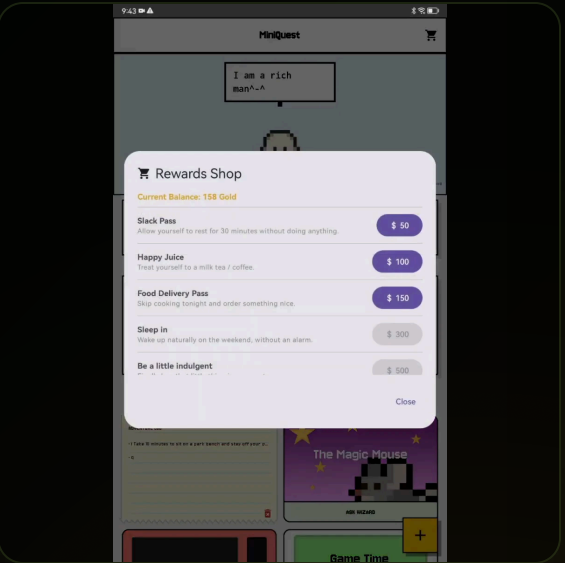
02 Level up

Confetti turns completion into an event.



03 Wizard's Hall

Guidance with online and offline fallback.



04 Rewards shop

Gold converts progress into chosen rewards.

Friendly enough to interrupt

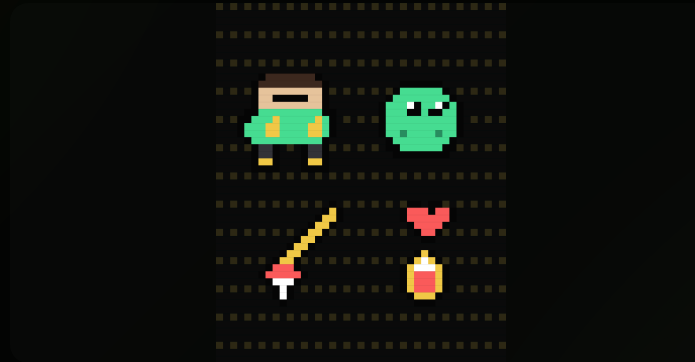
The visual language reduces the clinical feeling of intervention while keeping feedback crisp, immediate and legible.



01

RPG reward loop

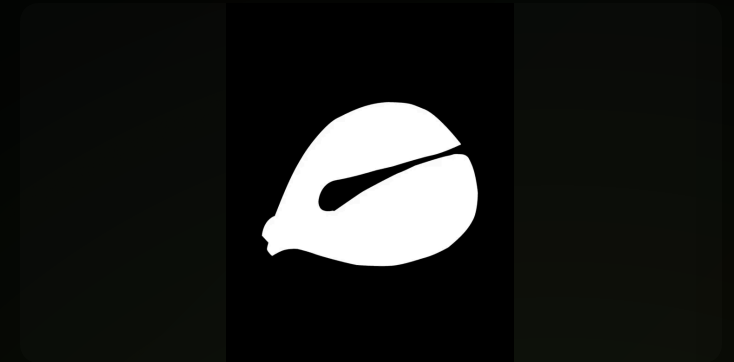
Habitica demonstrated that XP and gold can make self-management motivating; MiniQuest expands the loop beyond pre-written tasks.



02

Hard-edged pixel system

Vector pixel fonts and the custom HardShadowBox protect the playful 16-bit character on high-density displays.



03

One-tap regulation

The wooden-fish ritual informed a tactile stress-relief interaction that asks for less visual attention, not more.

Offline-first, reactive architecture

MVVM and unidirectional data flow keep sensor input, persistence, network calls and Compose UI separated but continuously synchronized.

VIEW

Jetpack Compose consumes immutable StateFlow and forwards user intent upward.

VIEWMODEL

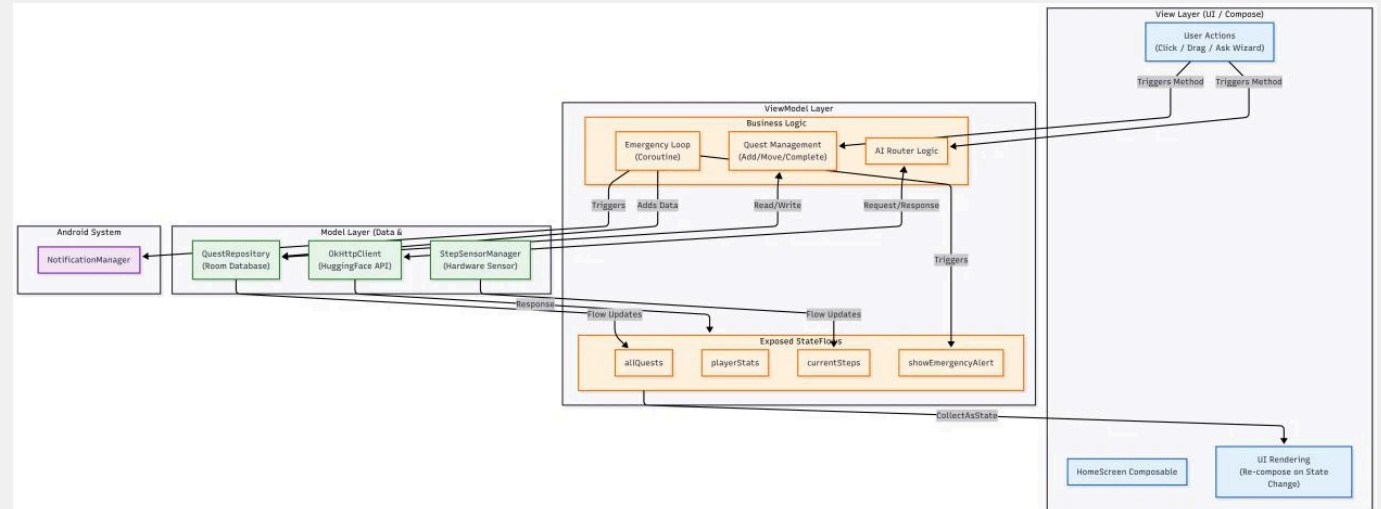
AppViewModel orchestrates repositories, sensors and AI calls inside lifecycle-safe coroutines.

MODEL

Room, StepSensorManager and OkHttp each retain one responsibility and remain UI-agnostic.

WHY IT MATTERS

Quests remain instant and usable offline; repository updates redraw the interface without manual refresh.



Submitted system architecture diagram

Engineering the experience

Two implementation decisions carry the concept: pervasive motion sensing and lifecycle-safe intelligence.

01 / SENSING

Raw acceleration, broader access

Using the tri-axis accelerometer avoids dependence on a dedicated step sensor that many lower-end devices lack.

StepSensorManager.kt

```
val magnitude = sqrt(x*x + y*y + z*z)

if (currentTime - lastShakeTime > 300) {
    internalSteps++
    saveStepsToStorage()
}
```

02 / ASYNC AI

A wizard that fails gracefully

viewModelScope cancels suspended requests with the screen; an offline grimoire keeps the interaction usable without a network.

AppViewModel.kt

```
viewModelScope.launch {
    _isWizardThinking.value = true
    val result = tryRouterModels(question)
    _wizardReply.value = result
    ?: offlineGrimoire.random()
    _isWizardThinking.value = false
}
```

Multimodal detail

Haptic confirmation reduces visual load, while vector pixel fonts and hard shadows preserve a crisp game identity.

What the work demonstrates

MiniQuest tests whether mobile engineering and behavioral design can reinforce each other rather than compete for attention.

01

Research translation

Converted two behavioral interventions into distinct product mechanics instead of adding theory after the interface was complete.

02

Mobile systems thinking

Connected sensing, persistence, reactive UI and network intelligence through a coherent offline-first architecture.

03

Experience craft

Used tactile feedback, celebration, pixel typography and role-play to make interruption feel inviting rather than punitive.

INDIVIDUAL CONTRIBUTION

Behavioral research, concept development, interaction design, visual direction, Kotlin implementation, testing and final curation.



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What remains unproven

This is a working Android prototype, not a claim of behavioural efficacy. Local Room data is lost on uninstall; phone-only acceleration cannot use wearables and is vulnerable to mechanical step-cheating.

Next validation

Firebase sync, Health Connect signals and ML-based waveform checks could improve continuity and sensing reliability. The central outcome is a tested technical system that makes interruption playful rather than punitive.